

the answer A based on the information contained in the passage P. This is known as the answer generation task.

Here is an example of the answer extraction task:

Passage P	Question Q	Answer A
Alyssa reached the beach after a long trip. She's from Charlotte . She travelled from Atlanta . She's now in Miami. She went to Miami to visit some friends. But she wanted some time to herself at the beach, so she went there first. After swimming in the ocean and lazing on the beach, she went to her friend Ellen's house. Ellen greeted Alyssa and both had some lemonade to drink. Alyssa called her friends Kristin and Rachel to meet at Ellen's house.	What city is Alyssa in?	Miami

The passage in the table above shows the relevant sentences marked for answering the question Q. Note that there are a number of locations/cities mentioned in the passage P. Hence, in order to correctly answer the question, "Which city is Alyssa in?" the program needs to make sense of the whole paragraph first before it can identify the relevant answer, which is, "She is now in Miami." From the relevant sentence, it should then extract the exact answer, namely 'Miami'. If you look carefully at the passage P, it does not mention the word 'city' anywhere. However, the question Q contains the word 'city'. When the program retrieves the answer as 'Miami', it needs to make sure that this is actually a city and not a state, county or continent. While named entity recognition can help classify Miami as the location, the program will need knowledge about the external world, including city names, to conclude that Miami is definitely a city.

Given that MRC is an important AI task, what are the data sets available for this problem? Prior to 2015, the only good data set available for the MRC task was the MCTest data set. This data set had only 2600 questions in it, which made it difficult to use deep learning approaches for this task. In recent years, a number of new MRC data sets have been released. These include the Stanford Question Answering Dataset (popularly known as SQuAD), the CNN/DailyMail QA data set, the NewsQA data set, and Microsoft's MS-MARCO data set. Of these, SQuAD is one of the most popular and a number of deep learning machine comprehension approaches have been applied to it.

SQuAD contains more than 200,000 questions with passages retrieved from Wikipedia articles. SQuAD's data set is a machine comprehension data set for the answer extraction task. You can explore SQuAD at <https://rajpurkar.github.io/SQuAD-explorer/>. The SQuAD leaderboard also shows the links to several of the techniques that have been applied to it. A popular data set for the answer generation variant of the MRC task is the MS-MARCO data set.

Now that we have understood the MRC problem, and the data sets that are available, let us quickly think about how we can approach this problem. Reading comprehension is a task done easily by most humans. We generally do a quick parse of the passage P first, then go over the question Q once, and again scan the passage P to find the relevant sentences to answer Q. We can design a simple baseline neural QA system to solve the MRC task based on a similar approach.

As we did in the duplicate question detection task, we first create a fixed length representation of question Q by passing it through a recurrent neural network such as LSTM. We also build a fixed length representation of passage P by passing it through another LSTM. We can then combine the two fixed length representations through an encoder. The combined representation is then used to predict the answer location in the passage. This simple approach is shown in Figure 1.

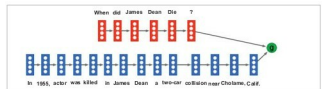


Figure 1: Predict the answer location in the passage

While this approach is conceptually simple and easy to understand, it has a number of drawbacks. By first encoding the question and the passage into fixed length representations, we compress the information contained in the text. This results in loss of information in matching the question to the relevant sentences which contain the answer in the passage P. One way of solving this problem is to adopt a two-stage approach. The first stage is to analyse the passage with the question Q to identify the key supporting fact in the passage P which can help answer the question. In the second stage, we build a representation of the passage where the key supporting fact is given the highest weightage among all the sentences contained in it. The representation of the encoded passage is then combined with the question and passed through the decoder, to predict the answer words present in the passage. This technique is known as 'memory networks' and we will discuss it in detail in next month's column.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Wishing all our readers happy coding until next month! 

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